

HS4002: Lecture 3B

Univariate Statistics

Ma Xiangyu

September 2, 2026



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Fetching lab notebooks

First, navigate to the right folder. Then, the following:

```
git checkout main  
git fetch upstream  
git merge upstream/main  
git push origin main  
git checkout -b week3
```

Setting up IDEs

Why Cursor?

- Cursor is an editor built on VS Code, with AI assistance added in.
- You can write Python, run Jupyter notebooks, and use the terminal in one place.
- You already have Python, Git, and the `hs4002` virtual environment from Week 1 — Cursor just needs to use them.
- JupyterLab still works. Cursor is the editor we will use from this week on.

Installing Cursor

1. Go to <https://cursor.com/download> and download the installer for your operating system.
2. Install it:
 - **Windows:** Run the installer. Keep the default options.
 - **macOS:** Open the downloaded file and drag **Cursor** into the **Applications** folder.
3. Open Cursor. Sign in with a GitHub account or an email address.
4. If you do not wish to use Cursor, you may also use Visual Studio Code – the two are very similar to each other.

Opening Your Lab Folder

1. Select **File** → **Open Folder**.
2. Choose your local `hs40021abs_2026` folder — the one you cloned in Week 2.
3. If asked whether you trust the authors of the files, select **Yes**.
4. Use the Explorer panel on the left to find this week's notebook, e.g. `labs/week3/week3_notebook.ipynb`.

Open the whole folder, not a single notebook. Cursor then finds your data files and the `.venv` sitting next to them.

Selecting the Python Kernel

1. Open a notebook ending in `.ipynb`.
2. Click **Select Kernel** in the top-right corner.
3. Choose **Python Environments**, then pick the interpreter inside the course `.venv`.
→ It may appear as Python (`.venv`) or Python (`hs4002`).

Running a Notebook in Cursor

1. Click the triangular **Run** button beside a cell, or press `Shift+Enter`.

2. Run the first cells and check that the imports work:

```
import pandas as pd
import numpy as np
import plotnine as p9
```

3. If a cell fails with `ModuleNotFoundError`, the wrong kernel is selected — go back and pick `.venv`.
4. Use the built-in terminal (**Terminal** → **New Terminal**) for git commands. Activate the venv there the same way you did in Week 1.

Turn Off Cursor Tab (For Now)

- Please **turn off Cursor Tab** for now — the grey AI suggestions that you accept by pressing Tab.
- You are still learning Python syntax and library names (pandas, plotnine, and so on). Typing code yourself builds fluency; accepting AI completions can hide what you do not yet know.
- In Cursor:
 1. Open **Cursor Settings** (gear icon in the top-right, or **Cursor** → **Settings** → **Cursor Settings** on macOS).
 2. Disable **Cursor Tab**.
- You can turn it back on later once you are more comfortable with the basics.

Type of Variables

Common Variables in Surveys

- Categorical variables, where the possible values are categories
 - Binary variables where there are two (yes/no) categories
 - Nominal variables where there are >2 categories, whose order does not matter.
 - Ordinal variables where the order of categories matters.
- Continuous variables, where the possible values are real numbers, and the intervals between them equally spaced.
- Different statistics and methods are useful for each of these.

Moments of a Distribution

What is a Moment?

The moments of a probability distribution are statistics that describe the shape of the probability distribution. They provide a systematic way of understanding how data is distributed around a central point.

We will look at four moments. You have already encountered many of them.

1. Mean
2. Variance
3. Skewness
4. Kurtosis

1st Moment: Mean

- Describes the central tendency of the distribution, its expected value.

2nd Moment: Variance

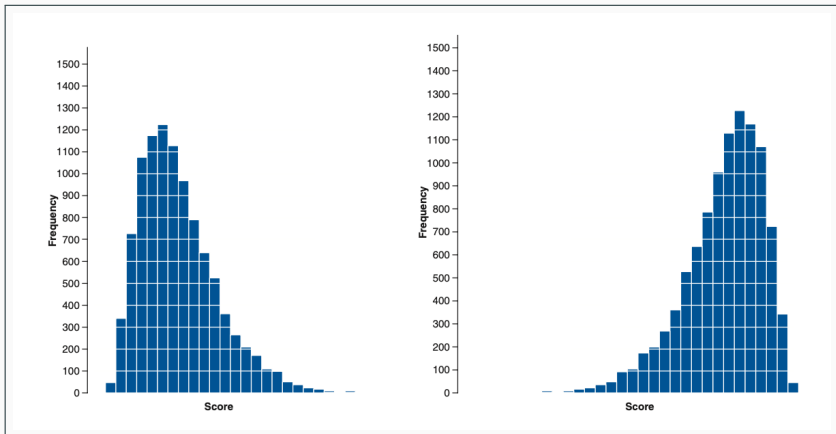
- Describes the dispersion or spread of the distribution around the mean.

3rd Moment: Skewness

- Describes the degree and direction of the asymmetry of the distribution around the mean.
- A symmetric distribution has a skewness of 0.
- $\text{Skew} > 0$ means it's positive-skewed.
 - Or, that the tail is on the right (i.e. a "longer" trail on the right).
- $\text{Skew} < 0$ means it's negative-skewed.
 - Or, that the tail is on the left (i.e. a "longer" trail on the left).

Visualizing skewness

Figure 1: Skewness (left figure is positively skewed; right figure is negatively skewed)

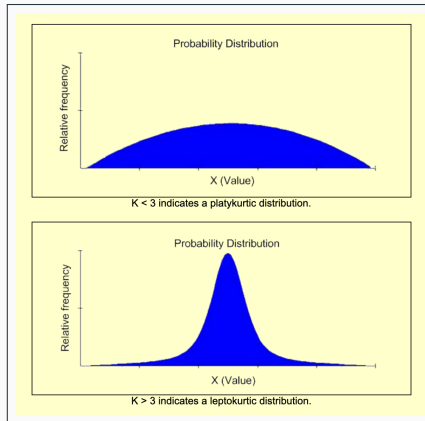


4th Moment: Kurtosis

- Kurtosis is a measure of the heaviness of the tails of a distribution.
- Mesokurtic distributions (kurtosis = 3) have the kurtosis of a normal distribution.
- Heavy tailed distributions will have kurtosis > 3 (leptokurtic; slim high peak with long tails)
- Light tailed distributions will have kurtosis < 3 (platykurtic; flatter and more short-tailed)

Visualizing kurtosis

Figure 2: top = platykurtic (<3); bottom = leptokurtic (>3)



Data Visualization for Univariate Statistics

- We use the histogram to visualize the frequency distribution of **continuous variables**.
- We use bar charts and tabulations to visualize the frequency of **categorical variables**.

Example of Kim and Brunn-Bevel (2023)

- Kim & Brunn-Bevel (2023) study the racialization of voice actor casting in popular animated films.
- It's an example of how you can produce interesting studies with "simple" statistics.

Figure 3: Most voice actors are white.

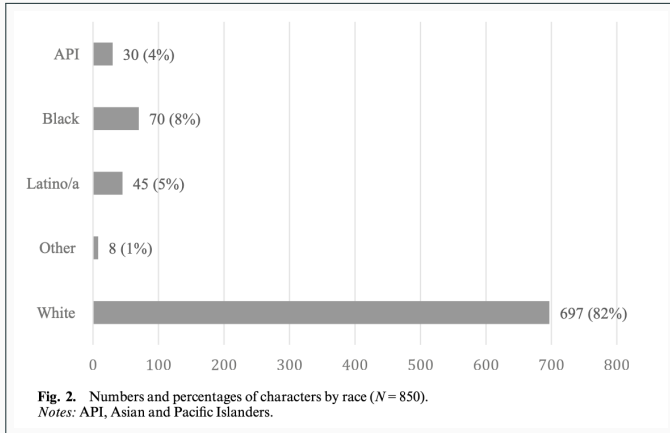
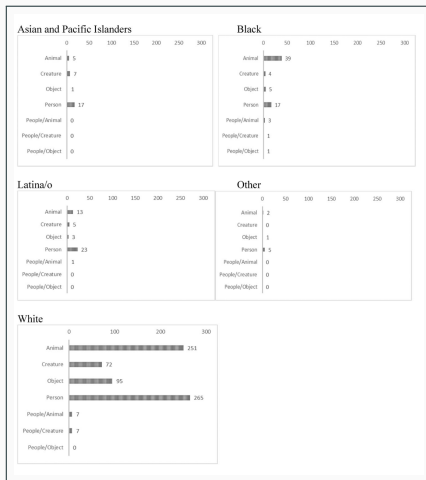


Figure 4: Black voice actors tend to be asked to voice animal characters.



Thanks for listening :)